NON-PHOTOREALISTIC RENDERING OF STYLISH, DISTINCTIVE FACES

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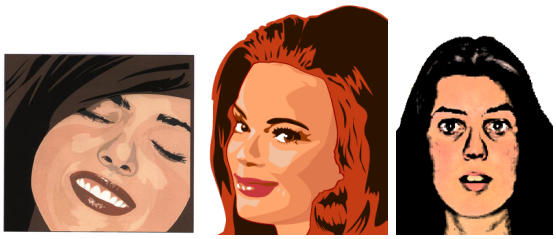
ABSTRACT

In this paper, we propose a novel non-photorealistic rendering algorithm that renders stylish, distinctive and colored faces. The rendered faces represent original faces with simplified colors and outlines of facial features. Meanwhile, the outlines of the distinctive features will be spotlighted with thicker lines. We extract the wavelet features of faces and exaggerate these features through a statistical approach. A parameterized histogram specification scheme is proposed to yield the final stylish faces. The proposed approach does not define any semantic facial feature, and it can spotlight any distinctive facial feature including beards, nevi or eyeglasses. Rendering of faces from the AR face database yields visually satisfying results.

Keywords: Non-Photorealistic Rendering, face caricaturing, face illustration

1. INTRODUCTION

Non-photorealistic rendering, or **NPR** for short, of faces emphasizes the funny and extraordinary characteristics of human faces. The rendering results can be in several forms: *sketches* which captures the facial edges, facial image *abstractions* and *cartoon* faces which represent faces using simplified edges, lines and colors, or *caricatures* that exaggerate distinctive facial features. Fig. 1 (a) and (b) illustrate two face drawn by artists. The goal of this paper is to render such face illustrations automatically, and one of our result is illustrated in Fig. 1 (c). NPR of faces is widely applied in newspapers, magazines, and some computer-human interfaces in which an avatar exists. Meanwhile, it is an alternative of a photorealistic picture when a portrait is needed (e.g. personal homepages, personal icons in MSN or Bulletin Board Systems).



(a) by Kristy Burks (b) by Seamus Leonard (c) our result

Figure 1: Face illustration examples.

Some NPR algorithms for faces produced sketches of faces by finding the outlines of facial features [13] [7] [6] [15]. Li and Kobatake proposed a method to extract facial sketches from facial image [13]. A generalized symmetry operator, rectangle-filter and a geometric template are used to detect eyes and mouth. Based on those locations, other facial parts are detected using their relative geometrical locations and their characteristic shapes. Then, feature points are determined from edges of each facial part, and line-drawings connecting them give the facial sketches. Hsu and Jain proposed to model facial components such as eyes, mouth, face outline, and the hair outline with a snake for each component [7]. The authors have demonstrated that the collection of the zero contours of these snakes can generate cartoon faces and caricatures. Authors in [6] and [15] have proposed algorithms which render black-and-white and colored illustrations of human faces, respectively.

Image abstraction simplifies the scene information in the image while retaining or emphasizing meaningful features. Kang and Lee proposed an image abstraction algorithm for producing stylish

abstraction of a photograph by preserving its directionality of features and shape boundaries [9]. It rendered interesting non-photorealistic faces, which are more visually pleasing than those rendered by conventional approaches such as image segmentation or quantization. A cartoon face is slightly different from a face abstraction. It typically represent faces with lines of specific facial components, such as eyes, nose and mouse. Works that rendered cartoon faces can be found in [7] [22] [21].

Face caricaturing can be regarded as another form of NPR, and it is a kind of exaggerative representation of faces which emphasizes the funny and extraordinary characteristics of human faces. Basically, a face caricature exaggerates the facial features which are distinctive compared with the average human face. Some algorithms create caricatures by learning the drawing styles of artists [2] [14] [3] [17] [11], while others create caricatures by exaggerating the distinctive facial features of faces, without the aid of artists [8] [20] [21] [19]. Kaneko and Meguro used a 3D wireframe model of head to locate several feature points of facial parts such as eyes, nose and mouth [8]. A PCA subspace is learned upon the coordinates of feature points, for each facial part. Facial exaggeration is realized by multiplying the projected coefficients in the PCA subspace by a magnification factor. Such a procedure typically renders faces with exaggerated *shapes* of individual facial parts. On the top of [8], Xu et al. [20] further exaggerated the relative positions between facial parts. [19] also exaggerated the shapes and positions of facial parts, and applied non-photorealistic rendering processes to generate a line-drawing sketch caricature. Besides 2-D caricatures, some works have rendered 3-D face caricatures [10] [5] [5] [18]. Furthermore, some interactive 3-D caricatures designing systems can be found in [1] [4].

Face sketches [13] [7] [6] [15] and abstractions [9] render stylish faces. However, these algorithms are not able to tell distinctiveness of an individual face. Hence they can not emphasize distinctive features. On the other hand, face caricaturing algorithms render human faces with their prominent features exaggerated. These prominent features, such as eyebrows, eyes, nose and mouth, of human faces are typically predefined. However, undefined features such as beard and nevus, or accessories like eyeglassed, will disappear in the rendering results [2] [14]. Meanwhile, these approaches require to locate these predefined local facial features, either manually [19] or automatically [8] [20]. The results are sensitive to the local feature alignment. Furthermore, most of these approaches produce grayscale, sketch versions of faces [13] [7] [6].

In this paper, we propose a new non-photorealistic rendering algorithm that renders stylish, distinctive and colored faces. Similar to face sketches and abstractions, the rendered faces represent original faces with simplified colors and outlines of facial features. Meanwhile, the outlines of the distinctive features will be spotlighted with stronger strokes (i.e. thicker lines). And this can be regarded as a form of caricaturing, since the conventional caricatures exaggerate the *shapes* of distinctive features, while the proposed algorithm exaggerates the *edges* of them. Lange and Kyuma had proposed a face caricaturing work [12], and it is similar to our algorithm. They performed the Wavelet Transform on grayscale facial images, and project the resulting features into a PCA subspace. By adjusting the coefficients in this subspace, they rendered exaggerated faces. However, the caricaturing results reported in [12] is neither prominent nor visually pleasing, due to the fact that they used grayscale images and the limitation in exaggeration degree owing to the image normalization. The proposed algorithm extends [12] in two ways: (1) It renders colored images instead of grayscale ones; (2) We propose a novel *Parameterized Histogram Specification* scheme, which allows large degree of exaggeration.

The contributions of this paper is summarized as the following: (1) We propose an non-photorealistic rendering algorithm which renders stylish faces with their distinctive features spotlighted. The proposed algorithm extends [12] and renders more visually pleasing results. (2) The proposed algorithm does not define any semantic facial feature, and it can spotlight any distinctive facial features, such as beards, nevus or even eyeglasses. (3) The proposed approach provides different degree of facial exaggeration. (4) We provide a brief yet comprehensive literature survey of non-photorealistic rendering of faces. The rest of the paper is organized as the following. Section 2 introduces the proposed system, and Section 3 illustrates its rendering results. We conclude our work in Section 4.

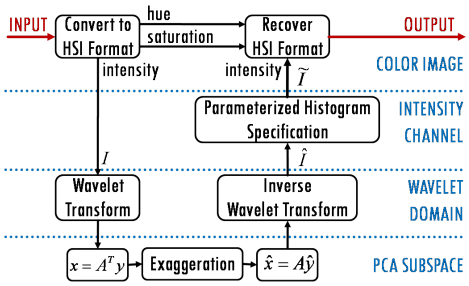
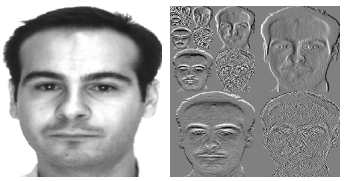


Figure 2: Overview of the proposed system.



(a) input (b) after wavelet transform

Figure 3: An example of 2-D wavelet transform.

2. THE PROPOSED SYSTEM

Diametric to the existing algorithms concerning NPR of faces, which locate the pre-defined local facial features and exaggerate their shapes, the proposed system aims to spotlight the distinctive facial features, which can be any facial parts, by giving them stronger edges. A 2-D wavelet transform on a face can detect horizontal, vertical and diagonal edges of different scales on that face. Fig. 3 illustrates an example of 2-D wavelet transform. The basic idea is to learn the distinctiveness and commonness among human faces in the wavelet domain using Principal Component Analysis (PCA). Meanwhile, the wavelet features are more robust to illumination variations [12]. Fig. 2 shows the proposed system architecture.

2.1. FEATURE EXTRACTION

Given a bunch of N colored training faces normalized by two eyes, we first project them into the HSI color space. We perform wavelet transform on the *intensities* channel. To do so, these grayscale images are resized to 2^n-by-2^n first (see Fig. 4 (b)). We use the filters $\frac{1}{2}(0,0,1,1,0,0)$ (low-pass, L) and $\frac{1}{8}(1,1,8,8,1,1)$ (high-pass, H) [12]. See Fig. 3 for an example of 2-D wavelet transform. For each of the resulting single-channeled 2-D image, we concatenate rows to form a 1-D feature vector, denoted as $\{x_1, x_2, ...x_N\}$, where N is the number of training faces.

(a) (b) (c) (d)

Figure 4: See text.

2.2. SUBSPACE LEARNING AND EXAGGERATION

In this paper, we exaggerate facial features in the manner similar to that in [12]. We perform PCA on x_i, $i = 1 \sim N$. The feature vector in the PCA subspace is denoted as y_i, and $y_i = A^T x_i$, where A is a projection matrix, each of its column is an eigenvector. These resulting eigenvectors record the major *edge patterns* of human faces. Let $\bar{y}$ denotes the mean vector over y_i, $i = 1 \sim n$.

To exaggerate a given face, first we extract its feature (see Section 2.1), denoted as x. Then, we obtain its projection in the PCA subspace, using $y = A^T x$. The exaggerated y, denoted as $\hat{y}$, is obtained using

$$\hat{y} = (1 + \alpha)(y - \bar{y}) + y. \quad (1)$$

The resulting exaggerated grayscale image, denoted as $\hat{I}$, is obtained by projecting $\hat{y}$ back to the wavelet domain followed by the inverse wavelet transform.

2.3. PARAMETERIZED HISTOGRAM SPECIFICATION

With different αs, different degrees of exaggeration can be obtained. If α is set large, the resulting intensities may be out of $[0, 255]$, [12] re-scaled the data to produce a 8-bit grayscale intensities. However, we empirically discovered that the facial exaggeration will be diminished if we re-scale the data linearly, no matter how large the α is set. To achieve larger exaggeration, we propose the *Parameterized Histogram Specification* scheme. This scheme also provides a way to control the degree of exaggeration.

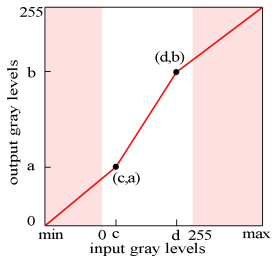


Figure 5: Illustration of the proposed Parameterized Histogram Specification.

First, we find the maximum and the minimum intensities among all $\hat{I}$s for a bunch of N training faces, and denote the two values as max and min, respectively. Based on the two values, the mapping function between the input and the output gray levels is illustrated in Fig. 5. In this figure, $\{a, b, c, d\}$ are the parameters that control the degree of exaggeration. Pixel locations with input gray levels within the *pink* regions are substantially *exaggerated* using equation 1, while those within the *white* regions are not. Basically $\{a, b, c, d\}$ can be arbitrarily assigned, as long as $c \leq d$. We propose a constrained model, which is

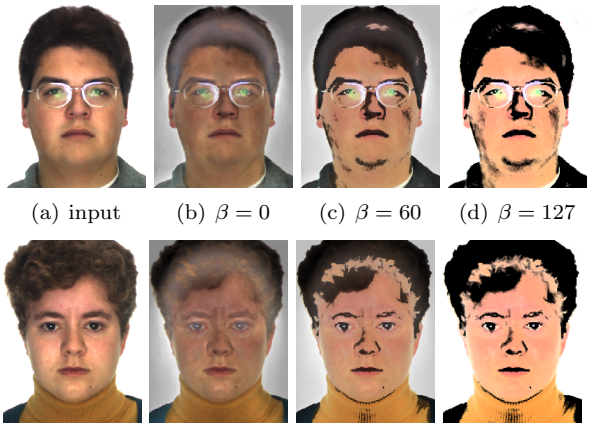
$$\begin{aligned} a &= c = 127 - \beta \\ b &= d = 127 + \beta, \end{aligned} \tag{2}$$

where $\beta \in [0, 127]$ is the parameter that determines the degree of facial *abstraction* and *exaggeration*. The mapping becomes linear when $\beta = 0$, and it is what was applied in [12]. Otherwise, this mapping is *piecewise linear*. Large β gives large degree of exaggeration, since in this case more distinctive edges and light-spots are generated.

Fig. 4 (c) illustrates an example of what is rendered with $\beta = 127$. The correspond colored image, which is obtained by adding the hue and saturation channels, is shown in Fig. 4 (d).

3. RENDERING RESULTS

We perform our experiments on the AR face database [16]. 252 frontal faces with the neutral expression from 126 identities are used to construct the PCA subspace and determine max and min for the Parameterized Histogram Specification scheme. Fig. 6 illustrates the rendering results using different β. We can see that a larger β renders a more exaggerated, stylish face. Fig. 7 gives more rendering results with $\beta = 127$. In Fig. 8, we compare our rendering result with those of other algorithms, including a basic color quantization scheme, a face sketching method [6], two face caricaturing works [12] [19], and an image abstraction algorithm [9], respecting the same person.[1] It is worthy of mention that [19] and the proposed algorithm both make the chin of this person sharper visually. We also include an interesting result, which is rendered using the proposed algorithm followed by [9], in Fig. 8 (h).



(a) input (b) $\beta = 0$ (c) $\beta = 60$ (d) $\beta = 127$

(e) input (f) $\beta = 0$ (g) $\beta = 60$ (h) $\beta = 127$

Figure 6: Rendering results under different β.

4. CONCLUSIONS AND FUTURE WORKS

We propose a new non-photorealistic rendering algorithm that renders stylish, distinctive faces. The rendered faces represent original faces with simplified colors and outlines of facial features. Meanwhile, the outlines of the distinctive features will be spotlighted with thicker lines. The proposed algorithm extends [12] and renders visually satisfactory faces.

The current algorithm may fail when the input faces are under significant out-of-plan rotations. Our future work will deal with this problem. One solutions is to create multiple PCA subspaces, each of which corresponds to a major face orientation. In this case, the algorithm will recognizes among

[1] We have implemented the color quantization scheme, [12], and [9]. The results of [6] and [19] are obtained from [19].

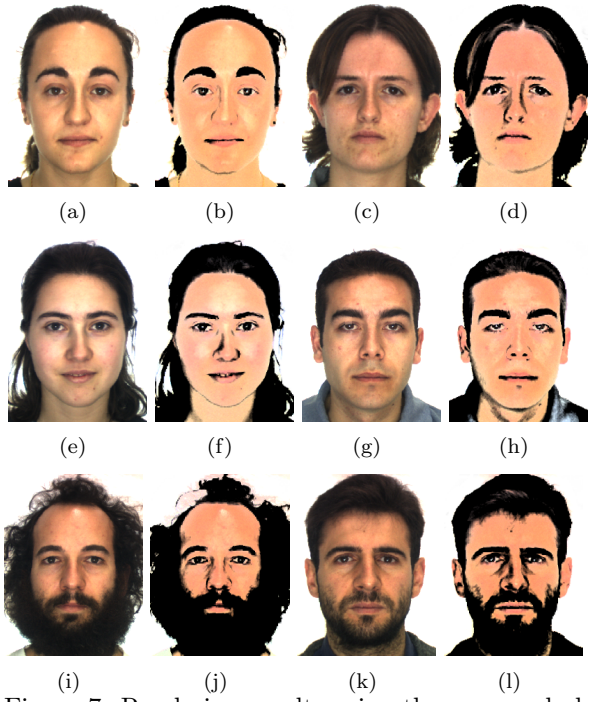


Figure 7: Rendering results using the proposed algorithm. Odd columns: input faces. Even columns: the corresponding rendering results.

these face orientations before the rendering.

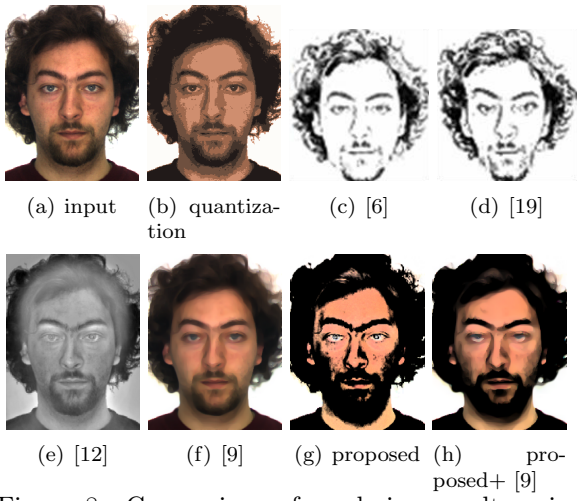


Figure 8: Comparison of rendering results using different algorithms.

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